LESSON PLAN

LSF, STABILITY, A/C PERFORMANCE (6)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	22
Total No. of Lesson Plans	33	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Turn forces in a turn, rate of turn steady level coordinated turn and control in a turn.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Turn forces in a turn, rate of turn steady level coordinated turn and control in a turn.
5. Specific Objectives	6.1.3.1 Trainee will be able to describe flight performance of aircraft.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of flight performance charts; Instructor supervision required for safety.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the primary forces acting on an aircraft in flight?
(b) Define a coordinated turn in terms of ball position.
(c) How does bank angle affect the load factor?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Centrifugal force in a turn • Load factor definition and calculation • Relationship between bank angle and load factor • Effect of load factor on stall speed • G-force sensation in high bank angles	L & Ds	PPT, IB	15 min
(b)	• Rate of turn formula derivation • Standard rate turn definition (3 degrees/sec) • Bank angle required for standard rate • Influence of airspeed on rate of turn • Turn coordination techniques	L & Ds	PPT, PDF	20 min
(c)	• Control inputs for coordinated turn • Aileron, rudder, and elevator coordination • Adverse yaw explanation and correction • Maintaining altitude during turn • Common pilot errors in turning flight	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how increasing bank angle affects the stall speed of an aircraft.
(b) Describe the control inputs required to maintain a coordinated standard rate turn.
(c) Why must a pilot apply opposite rudder when initiating a turn with ailerons?

12. Home Assignment and Follow Up	Prepare a brief note on how engine characteristics influence minimum speed and maximum rate of climb, linking these concepts to aircraft performance in LP 02.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Aircraft performance: a/c characteristics, engine Characteristics, minimum speed, maximum rate of climb, minimum Angle of climb, range and Endurance.	Refer to the master syllabus reference list	LP 23

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min